When mutualism weakens: stress alters endophyte benefits in cool-season grasses

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Abstract

Symbiotic mutualisms are widespread in nature and often thought to enhance host performance under environmental stress, yet the relative roles of abiotic and biotic factors in shaping these outcomes remain unclear. We studied how the independent and combined effects of aridity and herbivory modify the demographic effects of seed-transmitted fungal endophytes on three native grass host species using geographically distributed field experiments along a precipitation gradient, crossed with herbivore exclusion treatments. Host fitness declined under high aridity, with endophyte-symbiotic plants showing reduced survival, growth, or reproduction in at least one treatment for each species. Abiotic stress emerged as the primary driver of symbiosis outcomes, but in the opposite direction than predicted by theory and previous studies, with symbioses becoming more beneficial in more benign abiotic environments. These results highlight that symbiotic fungi can become liabilities under stress, with implications for the dynamics of plant-fungal mutualisms under global change.

Introduction

Symbiotic mutualisms are widespread and ecologically important, occurring across plant and animal hosts, and often mediating key ecological functions such as stress tolerance and defense against natural enemies. These interactions are famously context-dependent, with the direction and strength of outcomes shaped by the environment in which they occur (Bronstein, 1994; Hoang et al., 2024; Hoeksema and Bruna, 2015). The stress-gradient hypothesis (SGH) is a classic and influential framework for predicting the context-dependency of interaction outcomes, predicting that positive interactions increase under stressful conditions (Bertness and Callaway, 1994; Holmgren and Scheffer, 2010; Louthan et al., 2015). For example, microbial symbionts that benefit host plants under drought stress may become neutral or even parasitic under more benign conditions (Giauque et al., 2019).

As symbiotic interactions shift along stress gradients, their effects are predicted to become increasingly beneficial under biotic and abiotic stress, enhancing host resilience to environmental challenges. Under biotic stress, such as attack by natural enemies, symbionts can mediate defensive traits that reduce enemy performance, survival, or reproduction across diverse taxa (Atala et al., 2022; Bastias et al., 2017; Flórez et al., 2015; Haine, 2008; Vega, 2008). Similarly, under abiotic stress, including drought, nutrient limitation, and thermal extremes, symbiotic partners often enhance host tolerance through improved resource acquisition, physiological regulation, and stress-responsive metabolism (Afkhami and Rudgers, 2008; Cesar et al., 2024; Clay and Schardl, 2002). In plants specifically, microbial symbionts such as mycorrhizae, rhizobia, and endophytic bacteria and fungi can increase water and nutrient uptake, promote nitrogen fixation, and stimulate antioxidant production, allowing hosts to maintain growth and function under harsh conditions (Amijee et al., 1989). Many symbionts also stimulate the production of antioxidant enzymes, reducing oxidative damage caused by stressors such as drought, salinity, and heat (Ruiz-lazona et al., 1996). By improving resource acquisition, modulating plant physiology, and enhancing stress-responsive metabolism, these symbiotic interactions help host plants buffer against environmental stochasticity and extreme conditions across a wide range of ecosystems (Fowler et al., 2023; Li et al., 2026; Rudgers et al., 2020).

While stress is hypothesized to strengthen the benefits of symbiotic microbes, growing evidence suggests that it could also elevate costs of symbiosis. Maintaining symbionts requires that hosts allocate valuable carbon and nutrients to sustain their microbial partners. For instance, mycorrhizal fungi may require up to 20% of a host's photosynthetically fixed carbon (Soudzilovskaia et al., 2015; Thirkell et al., 2020), and under abiotic stress, this investment can outweigh the benefits of the symbiosis (Bronstein, 2001; Johnson et al., 1997), reducing host survival, growth, reproduction, or competitive ability (Klironomos, 2003). Similarly, vertically transmitted endophytes can shift from mutualistic to antagonistic interactions when they produce stromata that abort host inflorescences or impose energetic costs (Saikkonen et al., 2004). For example, under low nutrient or water conditions, endophyte-infected grasses may produce fewer tillers and lower total biomass compared to endophyte-free plants, reflecting the costs of sharing limited resources (Ahlholm et al., 2002). Thus, although stress is often expected to strengthen mutualistic outcomes, mounting evidence indicates that it can also intensify the demographic costs of hosting symbionts, raising fundamental questions about when symbioses enhance versus hinder host performance.

Despite growing interest in the role of symbiosis in shaping host demography, few studies have tested how abiotic and biotic stressors synergistically influence the demographic effects of symbionts. Furthermore, it remains unclear which of these stressors is more critical in eliciting the benefits and/or exacerbating costs of symbionts. Most work has been conducted in controlled. emphAmmophila breviligulata, drought did not intensify the effects of herbivory on growth, herbivory alone reduced performance, but endophyte-infected plants were more resistant to both stressors (Emery et al., 2010). Studying the combined effects of abiotic and biotic stressors under natural conditions is particularly important in the context of global change (Louthan et al., 2015), as shifting climate patterns and altered herbivore pressures may change the balance of costs and benefits in plant–microbe interactions, potentially influencing species distributions and ecosystem function (Reynolds et al., 2003). Understanding how abiotic stress, biotic stress, and microbial symbiosis interact to influence plant performance requires experiments that jointly vary all three factors. To our knowledge, such fully factorial studies remain rare outside controlled settings. Here, we address this gap by examining the effects of biotic and abiotic stress on the demographic effects of endophyte symbiosis on three grass host species spanning a natural range-core/range-edge precipitation gradient.

Symbioses between cool-season grasses and seed-transmitted fungal endophytes (*Epichloë* spp.) provide a powerful experimental model for studying how abiotic and biotic elements of ecological context influence host–symbiont interactions (Clay et al., 2005; Oberhofer et al., 2014; Saikkonen et al., 2010; Schardl et al., 2004). Because vertical transmission links host and symbiont fitness, it is expected to favor the evolution of mutualistic interactions (Gundel et al., 2011, 2024). Environmental conditions, including long-term variation in precipitation and temperature, strongly influence both the prevalence of *Epichloë* infection and the benefits conferred to hosts (Fowler et al., 2025; Rudgers et al., 2016). For example, grasses hosting endophytes often outperform endophyte-free individuals under water limitation, though the magnitude of these benefits depends on both host and symbiont identity (Decunta et al., 2021). A key benefit of *Epichloë* symbiosis is the production of fungal alkaloids that deter or reduce herbivore performance, thereby mediating plant–herbivore interactions (Clay et al., 2005;

Saikkonen et al., 2010). Hosting endophytes can also entail costs, including carbon allocation to sustain the fungus or risks associated with imperfect transmission. Finally, *Epichloë* endophyte effects can differ across host vital rates, which makes it important to consider multiple vital rates in assessing costs and benefits: symbiosis may not improve survival under stress, but can still enhance growth and reproduction (Rudgers et al., 2012; Yule et al., 2013).

We established experimental gardens of three species of native cool-season grasses across an aridity gradient in the south-central United States to investigate how endophyte symbiosis affects multiple host vital rates under varying abiotic and biotic stress. This gradient, from mesic conditions in the east to increasingly arid conditions in the west, provides a biologically realistic stress gradient, and coincides with the westernmost range limits of many eastern North American grass species. At each of seven sites along the gradient, we manipulated vertebrate and invertebrate herbivory (allowing herbivore access versus exclusion) to test its interaction with abiotic stress. Across these abiotic and biotic contexts, we quantified effects of symbiosis by comparing the demographic performance of endophyte-symbiotic (E+) and endophyte-free (E-) individuals. We asked whether endophyte symbiosis is beneficial for host grasses, whether symbiont benefits are amplified under exposure to abiotic or biotic stressors, as predicted by the SGH, and how these two forms of stress combine to influence the demographic outcomes of endophyte symbiosis. Specifically, we tested the following predictions (Fig. 1):

1. Endophyte symbiosis will generally be beneficial for the demography of these native grass hosts.
2. Because endophytes can improve drought tolerance (e.g., by enhancing water use efficiency or root growth), they will be most beneficial in arid environments, and neutral or costly in mesic environments.
3. Because endophytes produce defensive compounds (e.g., alkaloids), they will be most beneficial under herbivore exposure, and neutral or costly under herbivore exclusion.
4. The combined stresses of herbivory and low precipitation will amplify the physiological demands on the host plant, such that costs of maintaining endophyte associations will dampen net benefits relative to each stressor in isolation.

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[1]*One could predict the opposite, but there is limited evidence for "synergistic benefits" of multiple stressors in plant–microbe interactions, especially for endophytes. My logic follows Johnson et al. (1997) in New Phytologist, which shows that symbiotic associations can shift from mutualistic to parasitic when environmental conditions increase the costs of symbiosis relative to benefits (Johnson et al. 1997).*

Materials and methods

Study system

Our study focused on three cool-season perennial grass species native to woodland and prairie habitats of eastern North America: *Agrostis hyemalis*, *Elymus virginicus*, and *Poa autumnalis* (Poaceae, subfamily Pooideae) (Shaw, 2011). *A. hyemalis* is a short-lived perennial that germinates from autumn to late winter and, in our Texas study region, typically flowers between April and June. *E. virginicus* is a larger, longer-lived species that flowers throughout spring and summer. *P. autumnalis* has a similar growth form to *E. virginicus* and flowers in late spring. *Agrostis hyemalis* hosts the endophyte *Epichloë amarillans* (White Jr, 1994), *E. virginicus* typically hosts *Epichloë elymi* (Liu, 2023), and *P. autumnalis* hosts *Epichloë PauTG-1* or *E. typhina subsp. poae* (Gundel et al., 2020). Like other *Epichloë* species, these endophytes are primarily transmitted vertically through seeds and are additionally capable of horizontal transmission via sexual stromata on inflorescences or asexual conidia on leaf surfaces (Clay and Schardl, 2002; Gundel et al., 2020; Leuchtmann et al., 2014). In our study, stromata were never observed on *E. virginicus* or *Poa autumnalis* and very rarely observed on *A. hyemalis* (¡5% of individuals), suggesting low levels of horizontal transmission. Herbivory in this system is primarily from deer, generalist insects, and small mammalian grazers, which can influence both plant performance and endophyte-mediated defenses.

Source populations and symbiont removal

Plants used in our experiment were derived from 11 natural source populations located throughout the ranges of the focal species (Fig. 2, Table **??**). Seeds from these source populations (collected from ca. 30 plants per population) were either heat-treated or left untreated (control), intended to produce lineages of symbiont-free (E^{-}) and symbiotic (E^{+}) plants from the same genetic background. For *E. virginicus* and *P. autumnalis*, heat-treated seeds were placed in a drying oven at 60°C for five days. For *A. hyemalis*, seeds were heat-treated in a water bath at 60°C for 12 minutes. For all but two populations, the E^{-} plants used in our experiment were reared through one or more generations following heat treatment, which minimized the confounding potential of direct physiological effects of heating seeds (Table S2).

Seeds from control and heat-treated lineages were surface-sterilized with 10% bleach solution then germinated on water agar[2] plates. Seedlings were transferred to potting soil and grown in the Rice University greenhouse. Once these plants were sufficiently large they were

[2] *What percent agar?*

vegetatively propagated to help meet target sample sizes, with clonal identities tracked across individuals.[3] After vegetative propagation of *A. hyemalis*, *E. virginicus*, and *P. autumnalis*, the experiment included, respectively, 840, 840, and 720 plants (half E^+, half E^-) from three, four, and four source populations, with 92.3, 112.0, and 18.0 mean genotypes per source population and 5.47, 1.73, and 12.9 mean clonal replicates per genotype (Table **??**).

To assess the efficacy of heat treatment and to confirm endophyte status before transfer to the field experiment, we tested endophyte status (E^+ or E^-) of plants from control and heat lineages using three methods. First, for microscopy, peels of the inner leaf sheaths were stained with aniline blue lactic acid and viewed under a compound microscope at 200x–400x to identify *Epichloë* hyphae (Jackson and Johnson-Cicalese, 1988). Second, we used microscopy to assess the presence or absence of *Epichloë* in seeds that were squashed and stained similarly as the leaf tissue, to determine the endophyte status of the maternal plant (Fowler et al., 2025). Third, we used an immunoblot assay (Phytoscreen field tiller endophyte detection kit, Agrinostics Ltd. Co.) that targets proteins of *Epichloë* spp. to visually indicate presence or absence (Koh et al., 2006). In total, we tested 1,481 plants for endophyte status. In cases where two or more methods were used on the same plants ($N = 253$), there was 99.2% agreement between methods. For genotypes that were vegetatively propagated, we tested the endophyte status of a single clonal replicate and assume that its status applied equally to all clones.

All E^- plants used in the experiment described below came from heated-treated lineages, while E^+ plants came from both control and heat lineages depending on the efficacy of the removal treatment. While heat treatment was effective at reducing endophyte prevalence relative to controls, our confidence that individual E^- plants were experimentally disinfected, as opposed to naturally endophyte-free, varied due to among-population variation in endophyte prevalence (mean: 68% E^+) and the effectiveness of the heat treatment (Table S2). For example, for the SJC population of *A. hyemalis*, control and heat lineages yielded plants were 100% and 5% E^+, respectively – giving us high confidence that E^- plants are derived from successful symbiont removal. For the SFAEF population of *P. autumnalis*, on the other hand, control and heat lineages yielded plants were 35% and 26% E^+, respectively – meaning that these E^- plants were quite likely from naturally E^- lineages.

Common garden experimental design and climate data

To examine the effects of endophyte symbiosis across an abiotic stress gradient, we established common garden experiments with E+ and E- hosts at seven sites along an aridity gradient from Louisiana to central Texas, US (Fig. 2, Table S3). From west to east, mean annual precipitation

[3] *A reviewer may ask why genotype identity was not included as a random effect.*

increases more than three-fold across these sites while mean annual temperature is relatively consistent, based on climate data collected during the same period as our demographic surveys (Fig. S1). For this reason, we focus on precipitation as the primary abiotic factor that varied across sites. This geographic gradient overlaps and exceeds the natural range edges of our focal species, ensuring exposure to realistic climatic and biotic conditions (Fig. 2A,B, C).

To characterize climate at these sites during the study period (2023–2025), we collected monthly temperature and precipitation data from PRISM at 800 m resolution (Hart and Bell, 2015). For each site, we calculated mean temperature (°C) and cumulative precipitation (mm) over the corresponding census year, defined as the period from transplanting to the demographic census. During our study period, realized precipitation at these sites corresponded well to 30-year normals, with wetter eastern sites and drier western sites (Fig. 2D).

At each of seven sites along the gradient, we established 24 square plots (1.5 m $\times$ 1.5 m), eight per host species (*P. autumnalis* was planted at six sites, excluding the westernmost site (SON) since it is well outside this species' natural range: Fig. 2C). Within each site, each plot was randomly assigned to either herbivore access or herbivore exclusion (Fig. 2E). The herbivory treatment targeted both vertebrate and invertebrate hebivores: exclusion plots had deer fencing on four sides and were sprayed with Sevin insecticide (active ingredient Zeta-Cypermethrin) 2-3 times per growing season following the manufacturer's protocol; herbivore access plots were unsprayed and fenced on two sides to control for any fence effect.

Greenhouse-raised plants were planted into the field experiment in February-March 2023. Each plot was planted with 15 individuals, a mix of E+ and E- host-plants. Clonal replicates were arranged such that all plots had the same level of genetic diversity (number of unique genotypes) for each species. For each species at each site along the precipitation gradient, we planted a total of 60 E+ and 60 E- host-plants, half of those experiencing herbivore access and half experiencing herbivore exclusion. Plots at the two westernmost site (KER and SON) were destroyed by deer and subsequently replanted in 2024 with stronger fencing.

Demographic data collection

We collected demographic data, including survival, growth, and reproduction, during the flowering season of each species: *Agrostis hyemalis* and *Poa autumnalis* were censused in May and *Elymus virginicus* was censused in June of 2024, and 2025. For each individual, survival was recorded as alive or dead and, for surviving plants, size was measured as the count of tillers. Growth was quantified as the log ratio of tiller count between consecutive censuses (i.e., $\log(\text{tiller}_{t+1}/\text{tiller}_t)$) for plants that were alive in years t and $t+1$. We recorded the number of inflorescences per plant for all three species and counted the number of spikelets on up to three inflorescences from reproducing individuals of *Poa autumnalis* and *Elymus virginicus*. In *A. hyemalis*, one spikelet yields one seed, while in *P. autumnalis* and *E. virginicus* a single spikelet can produce 2-6 seeds. We measured herbivore damage to assess the effects of herbivory treatment on realized herbivory experienced by individual plants. We defined damage as chewing damage to leaves, tillers, or inflorescences, and estimated the proportion of damaged plants by plot, site, and herbivore treatment. These data showed that plants in fenced plots experienced less herbivory than in unfenced plots (Fig. 2F,Fig. S2,Fig. S3), confirming that our experimental manipulations had the intended effects.

Statistical modeling

To assess how abiotic and biotic stress modify the demographic effects of endophyte symbiosis, we fit generalized linear mixed models for each vital rate response in a hierarchical Bayesian statistical framework. All models shared the same hierarchical structure for their expected values and were applied to survival, growth, and fertility. Survival was modeled with a Bernoulli distribution, growth with a Gaussian distribution, inflorescence count with a zero-inflated negative binomial distribution, and spikelet count with a negative binomial distribution. All models included species-specific fixed effects for symbiosis (E+ or E-), precipitation (cumulative rainfall over 12 months preceding the census), and herbivory (herbivore access/exlusion), including all two- and three-way interactions. We considered models with linear and quadratic forms for the influence of precipitation, to account for the possibility of non-monotonic responses. Although we evaluated quadratic models, model selection criteria (LOOCV) showed that they provided minimal improvement over monotonic models (Table S4). Therefore, all subsequent analyses used the first-order (linear) effect of precipitation (Eq. 1), which captures the observed patterns while maintaining model simplicity. For each vital rate model, the expected value for the i^{th} observation (μ_i) was given by:

$$\begin{aligned}
\mu_i = {} & \beta_0[Sp_i] + \beta_P[Sp_i]precip_i + \beta_E[Sp_i]endo_i + \beta_H[Sp_i]herb_i \\
& + \beta_{EP}[Sp_i]endo_i precip_i + \beta_{HP}[Sp_i]herb_i precip_i \\
& + \beta_{HE}[Sp_i]herb_i endo_i + \beta_{HEP}[Sp_i]endo_i herb_i precip_i \\
& + \phi_{\text{site}[i],Sp_i} + \omega_{\text{source}[i]} + \rho_{\text{plot}[i]} \\
& \qquad \phi_{\text{site}[i],Sp_i} \sim N(0, \sigma^2_{site}) \\
& \qquad \rho_{\text{plot}[i],Sp_i} \sim N(0, \sigma^2_{plot}) \\
& \qquad \omega_{\text{source}[i],Sp_i} \sim N(0, \sigma^2_{source})
\end{aligned} \tag{1}$$

Here, $precip_i$ is total precipitation (mm) accumulated over the 12 months preceding the census, $endo_i$ indicates endophyte presence ($endo_i = 1$) or absence ($endo_i = 0$), and $herb_i$ indicates fencing treatment: herbivore access ($herb_i = 0$) or exclusion ($herb_i = 1$). The model includes random effects account for background variation at multiple hierarchical levels: $\phi_{\text{site}[i],Sp_i}$ captures site-to-site heterogeneity that is not explained by precipitation, $\rho_{\text{plot}[i]}$ captures plot-to-plot variation within sites, and $\omega_{\text{source}[i]}$ captures variation associated with the provenance of transplants (the latter two are unique to species and therefore not indexed by species). Random effect for site, plot, and source are host species-specific but the variance parameters (σ_{site}, σ_{source}, σ_{plot}) are shared across species; this assumes, for example, that site-to-site heterogeneity is of a similar magnitude across species, even as the model allows for "better" and "worse" sites to differ by species. This hierarchical structure allows the model to"borrow strength" across species (Gelman and Hill, 2007), improving estimates of fixed effects while still capturing species-specific responses to environmental gradient. Notice my change to the random effects in the equation. I think this is the more correct way to show plot and source effects, since they are already unique to species, and this should not require any changes to the code.

All models were implemented in R (R Core Team, 2023) using Stan via the `rstan` interface (Stan Development Team, 2024). Models were run with 4 chains and sufficient iterations to ensure convergence, with weakly informative priors specified for all fixed effects and zero-mean normal priors for random effects, whose standard deviations were estimated using weakly informative inverse-gamma priors. Full details on priors and model settings are provided in the accompanying code. We confirmed that models effectively captured key aspects of the grass demography data, including means, standard deviations, and quantiles using posterior predictive checks (Fig. S4).

We used posterior parameter estimates to visualize predictions for how effects of symbiosis varied across abiotic and biotic contexts. For each herbivory treatment (Access or Exclusion) and each precipitation level, we computed posterior predictions for the vital rates of E^+ and E^- hosts using the Bayesian model outputs (Eq. 1). From these predictions, we calculated the difference for each posterior sample ($\Delta = E^+ - E^-$) to estimate median effects and 90% credible intervals. We then computed the posterior probability that $\Delta > 0$, which quantifies the likelihood that endophytes enhance performance. A high posterior probability (e.g., > 0.9) indicates strong evidence of a beneficial effect, whereas a low posterior probability (e.g., < 0.1) indicates strong evidence of a cost (reduced performance). Probabilities near 0.5 indicate little or no consistent effect.

Comparing drivers of endophyte effects on host demography

To quantify relative statistical support for abiotic and/or biotic contexts as modifiers of grass-enodphyte symbiosis, we fitted four hierarchical Bayesian candidate models for each vital rate (survival, growth, inflorescence, and spikelet production; Supplementary Methods S.1.1). All models included endophyte status as a baseline predictor but differed in whether they incorporated interactions with precipitation (Endophyte $\times$ Precip), herbivory (Endophyte $\times$ Herbivory), both factors including the three-way interaction (Endophyte $\times$ Precip $\times$ Herbivory), or only main effects (Endophyte + Precip + Herbivory). We compared candidate models using leave-one-out cross-validation (LOO-CV) (Vehtari et al., 2017), which estimates predictive accuracy by iteratively holding out each observation. Differences in expected log predictive density (ΔELPD), along with associated standard errors, were used to rank model performance. When the difference in ELPD between two models (ΔELPD) was small relative to its standard error (SE), we considered their predictive performance to be effectively similar. Accordingly, model rankings were interpreted cautiously, focusing on consistent trends across vital rates rather than on small differences between closely ranked models.

Results

Endophyte effects on host fitness components across abiotic and biotic stress gradients

Wetter conditions increased host survival across all species, and abiotic stress modified endophyte effects on survival for two of the three grasses, but in the opposite direction as predicted: grass-*Epichloë* symbiosis generally had negative or neutral effects under low precipitation, but benefits increased under wetter conditions (Fig. 3; Fig. 6). Under herbivore access, *A. hyemalis* and *E. virginicus* showed median $\Delta(E^+ - E^-)$ values from -0.03 to 0.03 at low precipitation ($\Pr(\Delta > 0) = 0.19$–0.66; Table S5), increasing to 0.08–0.04 under high precipitation ($\Pr(\Delta > 0) = 0.63$–0.84; Table S5). Herbivore exclusion strengthened the switch from negative to positive effects. In *A. hyemalis*, Δ shifted from slightly negative at low precipitation to moderately positive at high precipitation ($\Pr(\Delta > 0) = 0.05$–0.86; Table S5). *E. virginicus* changed from negative under dry conditions to positive under wetter conditions ($\Pr(\Delta > 0) = 0.04$–0.97). *P. autumnalis* showed minimal survival benefits under herbivore access ($\Delta = -0.06$ to 0.00; $\Pr(\Delta > 0) = 0.15$–0.53; Table S5) and weakly positive effects under herbivore exclusion ($\Delta = 0.00$–0.03; $\Pr(\Delta > 0 = 0.62$–0.71; Table S5).

Endophyte effects on growth varied across species, increasing with precipitation in some cases and depending on herbivore access (Fig. 4; Fig. 6). In *A. hyemalis*, endophyte effects on growth were neutral under low precipitation when herbivores were present ($\Delta = 0.01$; $\Pr(\Delta > 0) = 0.51$;Table S6)), but became increasingly positive at moderate-to-high precipitation ($\Delta = 0.34$–0.69; $\Pr(\Delta > 0) = 0.98$–0.99; Table S6)). Under herbivore exclusion, growth benefits were negative to slightly neutral at low precipitation ($\Delta = -0.20$; $\Pr(\Delta > 0) = 0.26$) but increased with precipitation ($\Delta = 0.14$–0.51; $\Pr(\Delta > 0) = 0.84$–0.94; Table S6)). *E. virginicus* showed positive endophyte effects under herbivore access, with median Δ decreasing slightly from moderate to high precipitation ($\Delta = 0.55$–0.08; $\Pr(\Delta > 0) = 0.94$–0.63; Table S6)). Effects under herbivore exclusion were smaller and more uncertain, ranging from slightly negative at low precipitation ($\Delta = -0.09$; $\Pr(\Delta > 0) = 0.39$; Table S6)) to modestly positive at high precipitation ($\Delta = 0.17$; $\Pr(\Delta > 0) = 0.83$; Table S6)). In *P. autumnalis*, endophyte effects were generally strong. Under herbivore access, median Δ ranged from slightly negative at low precipitation to small positive values at high precipitation ($\Delta = -0.39$ to 0.24; $\Pr(\Delta > 0) = 0.19$–0.80;Table S6)). Under herbivore exclusion, endophyte benefits in *P. autumnalis* were largest at low-to-moderate precipitation ($\Delta = 0.47$–0.74, $\Pr(\Delta > 0) = 0.95$–0.97; Table S6)) but declined at high precipitation, becoming negative at the highest level ($\Delta = -0.60$, $\Pr(\Delta > 0) = 0.02$; Table S6)).

Endophyte effects on reproduction were generally neutral at low precipitation and became increasingly positive under wetter conditions, with benefits often slightly stronger under herbivore exclusion (Figs. 5,Figs. S5; Fig. 6;Tables S7, S8). In *A. hyemalis*, median Δ shifted from neutral or slightly negative at low precipitation to positive at moderate-to-high precipitation under both herbivore treatments. In *E. virginicus*, effects were weak or slightly negative at low precipitation under exclusion, becoming modestly positive at higher precipitation. In *P. autumnalis*, endophyte effects increased with precipitation, with strongest benefits under exclusion at low-to-moderate precipitation before declining at the highest level.

Abiotic factors as the primary drivers of fitness components across the stress gradient

The outcomes of grass-endophyte symbioses, whether beneficial, costly, or neutral, were primarily associated with climatic variation rather than herbivore treatment (Table 1). Across all vital rates (survival, growth, inflorescence, and spikelet production) and host species, model comparisons consistently indicated that models including an *Endophyte* $\times$ *Climate* interaction had the strongest predictive support. Although differences in expected log predictive density (ELPD) were modest relative to their standard errors for some traits, these patterns were consistent across all fitness components. For survival, the *Endophyte* $\times$ *Climate* model was clearly favored over models including *Endophyte* × *Herbivory* (ΔELPD = −0.33, SE = 2.60), the three-way interaction (ΔELPD = −1.92, SE = 1.43), or the additive Endophyte + Climate + Herbivory model (ΔELPD = −4.98, SE = 3.24). Similar patterns were observed for growth, inflorescence, and spikelet production, with the *Endophyte* × *Climate* model consistently ranking highest, though ELPD differences relative to the other models were smaller for reproduction (Table 1).

Discussion

Drawing on three years of field experiments and Bayesian demographic models, we evaluated how biotic and abiotic stressors jointly influence the demographic effects of vertically transmitted *Epichloë* endophytes on three cool-season perennial grasses across a natural climatic gradient in the southern United States. We predicted that endophytes would enhance host fitness in arid environments and in the absence of herbivores, representing an endophyte-driven stress amelioration scenario (Fig. 1A), and that the addition of herbivory would reduce or reverse these benefits due to compounded metabolic costs (Fig. 1B). Contrary to these expectations, we found little evidence that endophytes buffered hosts against declining precipitation, even when herbivores were excluded. Instead, endophyte-mediated benefits were strongest under relatively benign, mesic conditions and weakened as abiotic stress intensified toward the arid range edges of these host species. Model comparisons confirmed that the effects of endophytes on host fitness depended on climate, whereas herbivory had a weaker and less consistent influence (Table 1). These patterns do not support SGH-based predictions and instead suggest that increasing abiotic stress constrains both host performance and the capacity of endophytes to confer demographic benefits.

Endophyte presence is often assumed to enhance host performance under favorable environmental conditions, with some studies reporting increased growth, reproduction, or survival in grass-*Epichloë* systems (Clay, 1988, 1990; Gibert et al., 2012; Rodriguez et al., 2009). Surprisingly, in our study, endophyte benefits declined under low-precipitation conditions, even though drought mitigation is sometimes reported for these symbioses. This unexpected pattern suggests that environmental stress may alter the costs and benefits of the symbiosis, potentially limiting the advantages that endophytes provide under harsher conditions. Reduced water availability may limit nutrient uptake and increase the metabolic costs of maintaining the symbiosis, tipping the cost-benefit balance against the host (Ahlholm et al., 2002; Bronstein, 2001; Faeth, 2002). One possible mechanism is that severe drought can activate plant defense pathways, including elevated phytohormone production and reactive oxygen species, which may inhibit endophyte growth or reduce its functional contributions to the host (Bastías and Gundel, 2023; Eaton et al., 2011). Interestingly, these effects were observed even without herbivores, suggesting that abiotic stress alone can weaken the host-endophyte relationship. This contrasts with studies emphasizing herbivore-mediated context dependence (Afkhami and Rudgers, 2009; Bastias et al., 2017), indicating that environmental limitations can independently generate similar outcomes.

Drought simultaneously restricts water and nutrient availability while increasing metabolic demands; thus, the host's carbon budget may be a key factor shaping the net costs and benefits of symbiosis. Water-stressed plants close stomata to conserve water, reducing photosynthetic carbon gain while still expending C to maintain cellular homeostasis (Dietze et al., 2014). Under these conditions, sustaining an endophyte may represent a relatively high carbon cost. Similar to other nutritional mutualisms, such as nitrogen-fixing rhizobia, water stress could reduce C allocation to the symbiont, although microbial partners may partially offset these costs by enhancing photosynthesis or water-use efficiency (Pringle, 2016; Toby Kiers et al., 2010). While carbon is a primary integrative currency, other factors including nutrient limitation and shifts in plant hormones also likely influence symbiosis outcomes. Future studies could quantify these costs by measuring nonstructural carbohydrates (NSCs) across tissues and over time along precipitation gradients, providing a clearer picture of how carbon limitation, in combination with other stressors, shapes the net outcome of endophyte-host interactions. Such approaches would improve understanding of the energetic balance of these interactions under varying abiotic conditions and help predict how environmental change may shift mutualism outcomes, with implications for both ecological theory and the management of plant-microbe systems.

Abiotic stress was the primary driver of endophyte-mediated effects across species and fitness components, while herbivory played a secondary role that depended on site-specific herbivore pressure. Water limitation strongly influenced host demography in our system, with drier sites constraining survival, growth, and fertility. Contrary to predictions of the stress gradient hypothesis (SGH), which posits that biotic interactions should become more positive under stressful conditions (Lortie and Callaway, 2006; Louthan et al., 2015), endophyte-mediated benefits were reduced under high abiotic stress. Most empirical tests of SGH have focused on plant–plant interactions (Bertness and Callaway, 1994), and while some recent syntheses suggest microbial interactions may show stronger support for SGH (Hammarlund and Harcombe, 2019; Piccardi et al., 2019), our results indicate that host-associated plant microbe mutualisms may not conform to this pattern (David et al., 2018; Hernandez et al., 2021). Unlike neighboring plants, which can reduce environmental stress for themselves and others for example, by shading soil, retaining moisture, or improving nutrient availability endophytes live entirely within their host and cannot directly modify the external environment. As a result, the benefits they provide depend on how well the host is coping with stress. When survival, growth, or reproduction is strongly limited by drought, the endophyte's capacity to enhance host fitness may be reduced.

Our study demonstrates that variation in precipitation across natural climatic gradients directly shapes host-endophyte interactions. By linking the demographic consequences of endophytes to environmental gradients, we take an important step toward forecasting the resilience of plant–microbe interactions under ongoing environmental change. These results align with broader evidence that the benefits of microbial mutualisms are influenced by environmental conditions. Recent work shows that microbial mutualisms can buffer hosts against drought and support population persistence, but their effectiveness is highly context-dependent (Li et al., 2026). Likewise, historical analyses of Epichloë prevalence across multiple grass species indicate that long-term changes in precipitation and temperature have already influenced symbiont distributions, highlighting the potential for environmental shifts to reshape mutualistic outcomes over time (Fowler et al., 2025). Moving forward, quantifying how symbioses influence population growth under different climate-change scenarios including increased drought will be critical to predicting whether mutualisms are likely to be disrupted or strengthened across host ranges, and to understanding the implications for species distributions.

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Figures

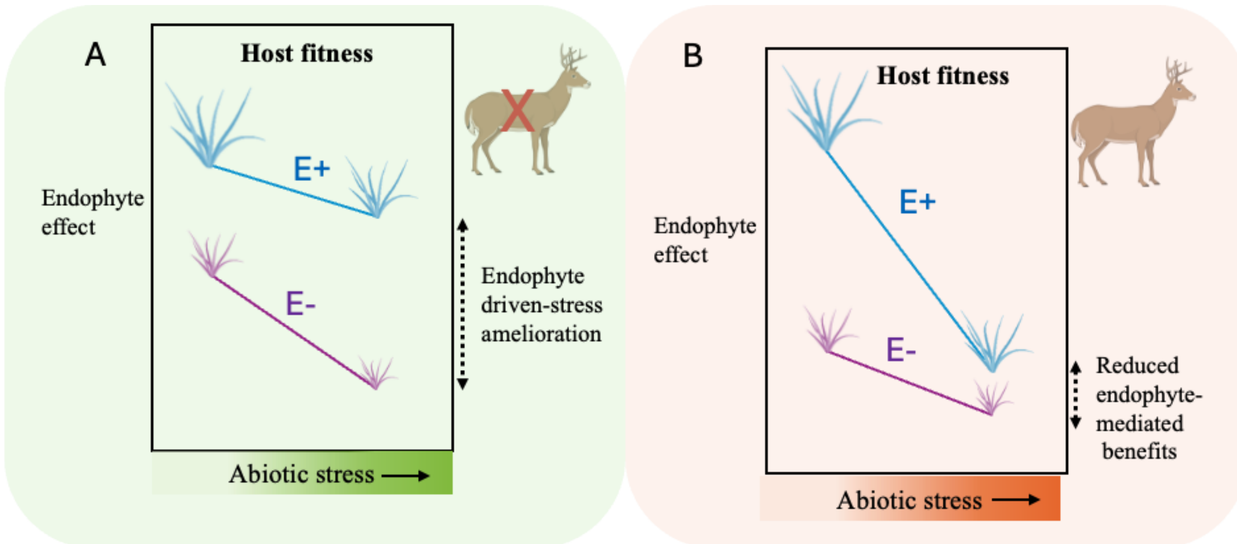

Figure 1: Conceptual framework illustrating predicted effects of abiotic stress, biotic stress, and endophyte-mediated mutualism on host plant fitness (adapted from (Porter et al., 2020)). (A) Endophyte-driven stress amelioration: endophytes enhance host performance under increasing abiotic stress in the absence of herbivores. (B) Reduced endophyte benefits under combined abiotic stress and herbivory due to resource limitation. Blue lines represent endophyte-infected plants (E^+) and violet lines represent endophyte-free plants (E^-).

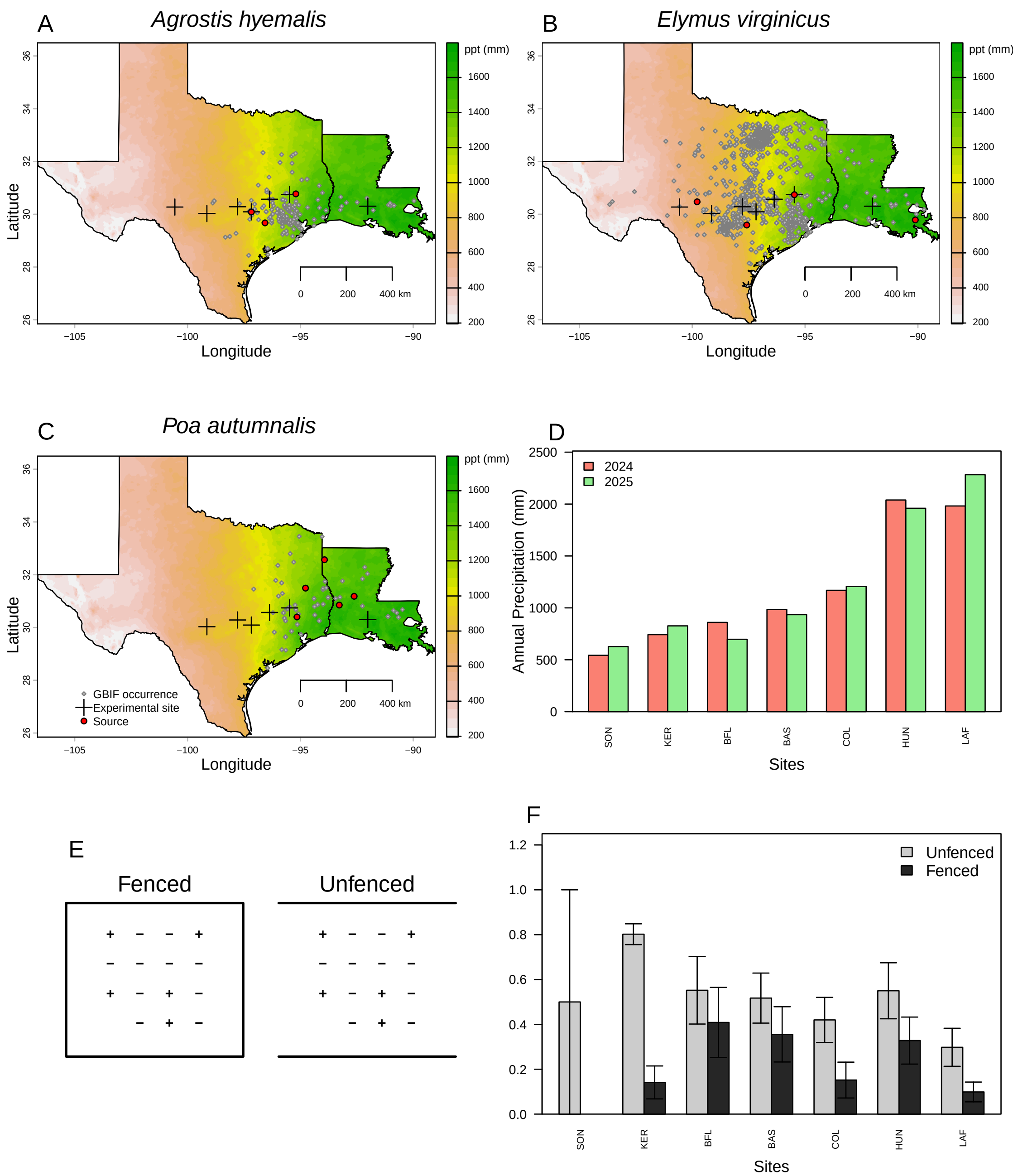

Figure 2: Distribution of common garden sites along an aridity gradient from Texas to Louisiana, USA, for (A) *Agrostis hyemalis*, (B) *Elymus virginicus*, and (C) *Poa autumnalis*. Panels show PRISM 30-year climate normals (1996–2025). Red dots indicate source populations and grey dots GBIF occurrences. (D) Annual precipitation during census years. (E) Herbivory treatments (access vs. exclusion). (F) Proportion of damaged plants (mean ± SE).

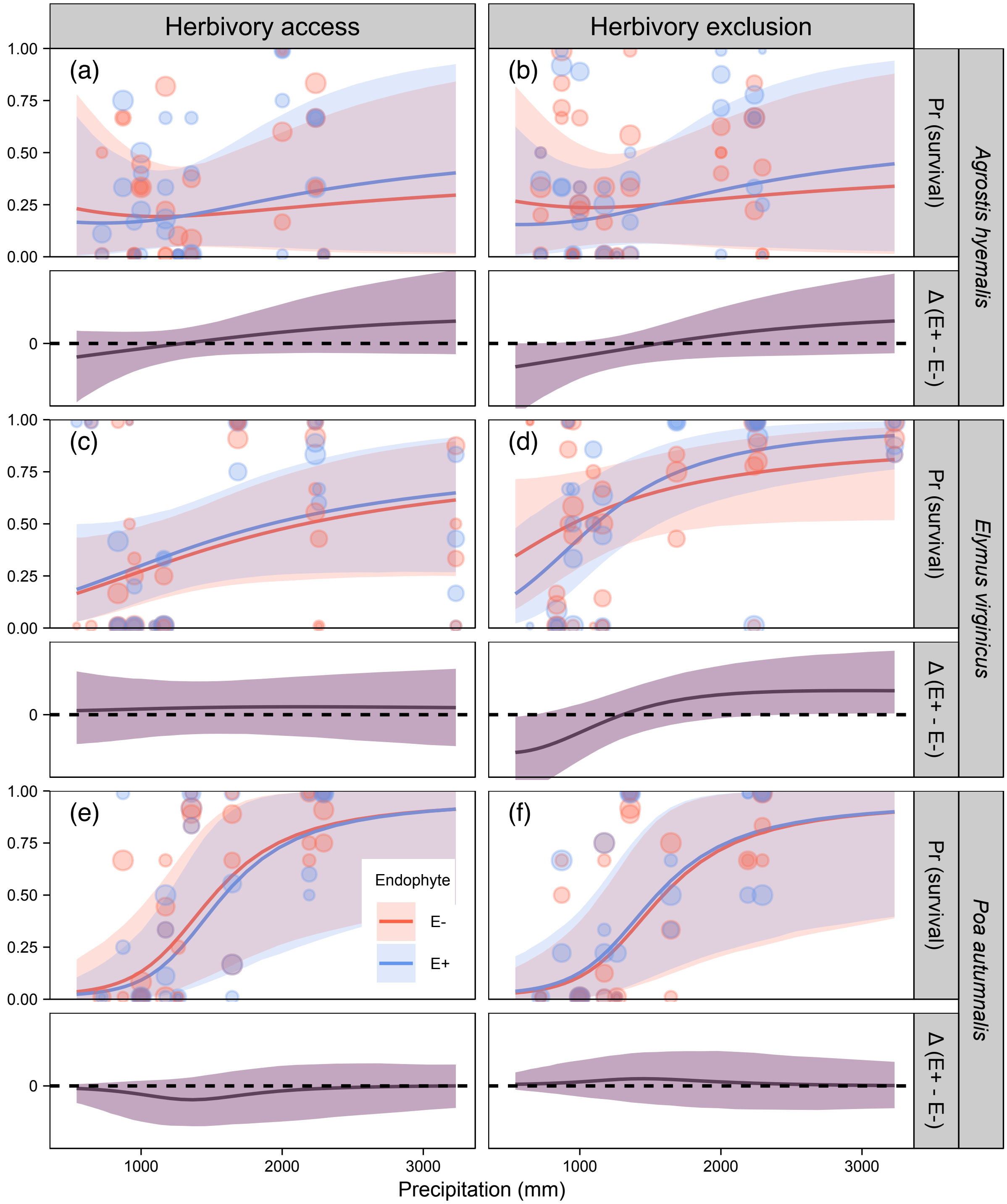

Figure 3: Predicted survival probability (upper panels) and endophyte effect ($\Delta = E^{+} - E^{-}$; lower panels) across precipitation gradients. Lines show posterior means with 90% credible intervals. Points indicate observed means, with point size proportional to sample size. Horizontal dashed line denotes $\Delta = 0$. Panels are arranged by species and herbivory treatment (herbivore access and exclusion).

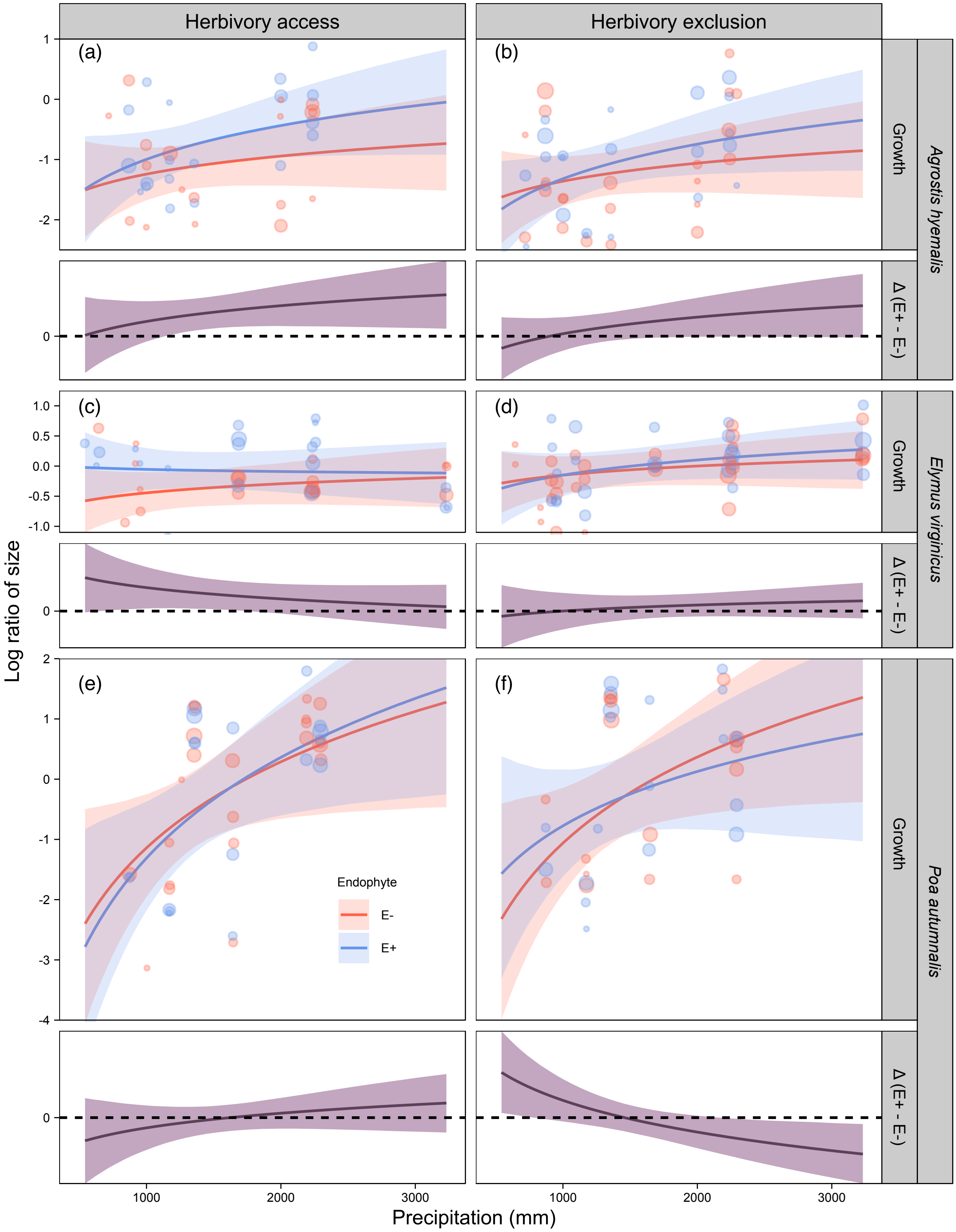

Figure 4: Predicted relative growth rate (upper panels) and endophyte effect ($\Delta = E^{+} - E^{-}$; lower panels) across precipitation gradients. Lines show posterior means with 90% credible intervals. Points indicate observed means, with point size proportional to sample size. Horizontal dashed line denotes $\Delta = 0$. Panels are arranged by species and herbivory treatment (herbivore access and exclusion).

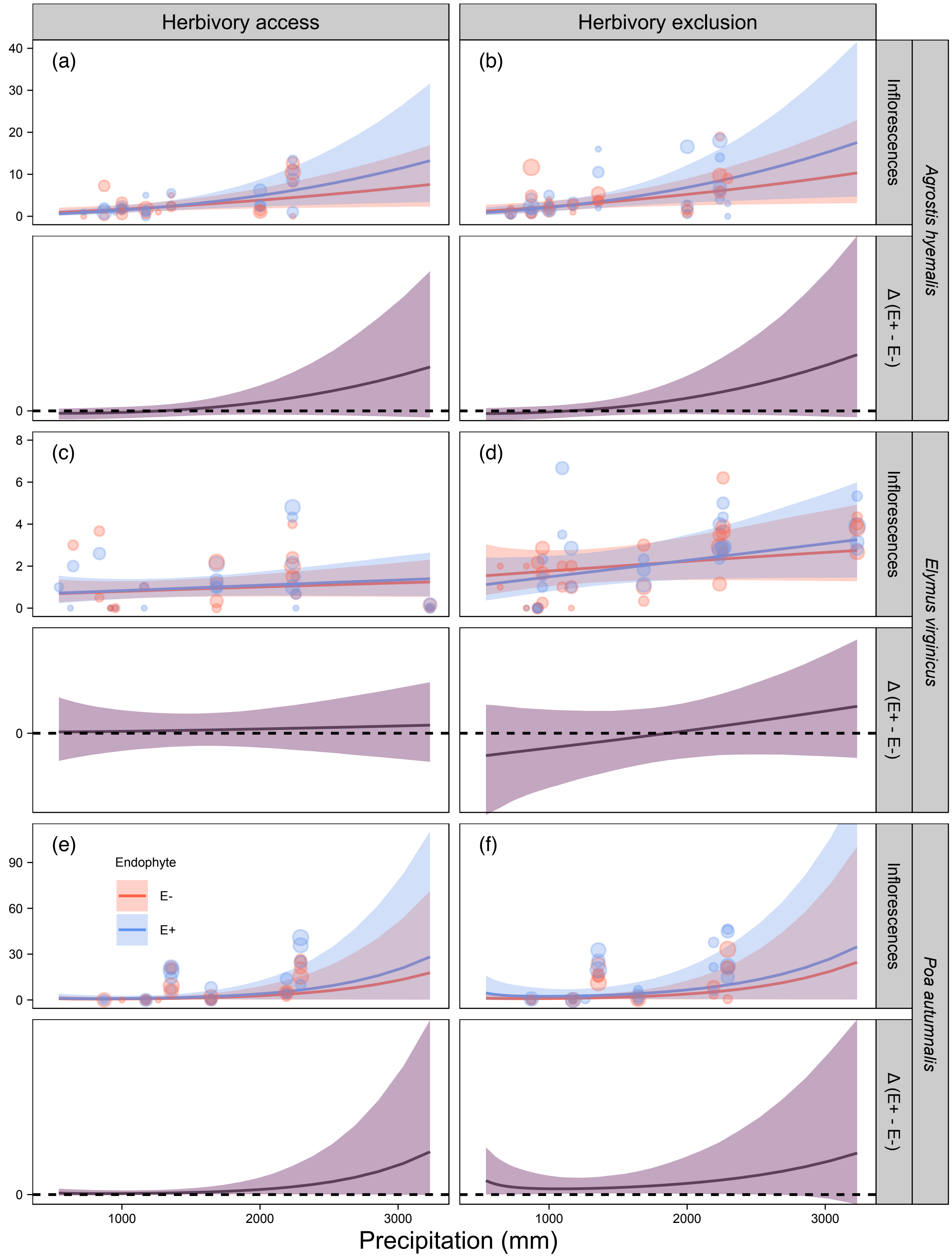

Figure 5: Predicted inflorescence production (upper panels) and endophyte effect ($\Delta = E^{+} - E^{-}$; lower panels) across precipitation gradients. Lines show posterior means with 90% credible intervals. Points indicate observed means, with point size proportional to sample size. Horizontal dashed line denotes $\Delta = 0$. Panels are arranged by species and herbivory treatment (herbivore access and exclusion).

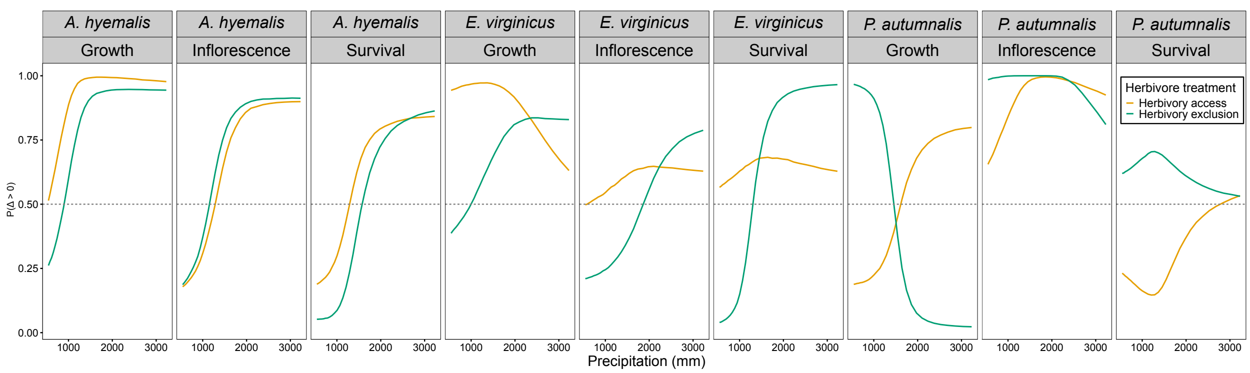

Figure 6: Predicted effects of herbivory exclusion on survival, growth, and inflorescence production across precipitation gradients. Panels show median differences and $\Pr(\Delta > 0)$. Horizontal dashed lines denote $\Delta = 0$ and $\Pr(\Delta > 0) = 0.5$. Panels are arranged by species and vital rate.

Tables

Table 1: Comparison of models assessing the drivers of grass–endophyte symbiosis across four fitness components (vital rates). ΔELPD represents the difference in expected log predictive density relative to the best-fitting model, and SE denotes the standard error of this difference.

Vital rate	**Model type**	**ΔELPD**	**SE**
Survival	Endophyte × Climate	0.00	0.00
	Endophyte × Herbivory	−0.33	2.60
	Endophyte × Climate × Herbivory	−1.92	1.43
	Endophyte + Climate + Herbivory	−4.98	3.24
Growth	Endophyte × Climate	0.00	0.00
	Endophyte × Herbivory	−1.72	1.99
	Endophyte × Climate × Herbivory	−2.04	3.05
	Endophyte + Climate + Herbivory	−2.47	3.72
Inflorescence	Endophyte × Climate	0.00	0.00
	Endophyte × Herbivory	−1.65	1.81
	Endophyte × Climate × Herbivory	−3.02	2.65
	Endophyte + Climate + Herbivory	−4.65	2.70
Spikelet	Endophyte × Climate	0.00	0.00
	Endophyte × Herbivory	−0.56	2.25
	Endophyte × Climate × Herbivory	−0.95	1.54
	Endophyte + Climate + Herbivory	−4.23	1.87

S.1 Supporting Information

S.1.1 Supporting Methods

Candidate models for abiotic and biotic drivers of cool-season grass species

1. **Endophyte × Climate model:** includes endophyte status (endo), precipitation (P), and their interaction:

$$\mu_i = \beta_0[\text{Sp}_i] + \beta_{\text{endo}}[\text{Sp}_i] endo_i + \beta_P[\text{Sp}_i] P_i + \beta_{\text{endo}\times P}[\text{Sp}_i] endo_i P_i \\ + \phi_{\text{site}[i],\text{Sp}_i} + \omega_{\text{source}[i],\text{Sp}_i} + \rho_{\text{plot}[i],\text{Sp}_i}$$

2. **Endophyte × Herbivory model:** includes endophyte status (endo), herbivory (herb), and their interaction:

$$\mu_i = \beta_0[\text{Sp}_i] + \beta_{\text{endo}}[\text{Sp}_i] endo_i + \beta_{\text{herb}}[\text{Sp}_i] herb_i + \beta_{\text{endo}\times\text{herb}}[\text{Sp}_i] endo_i herb_i \\ + \phi_{\text{site}[i],\text{Sp}_i} + \omega_{\text{source}[i],\text{Sp}_i} + \rho_{\text{plot}[i],\text{Sp}_i}$$

3. **Endophyte × Climate × Herbivory:** combines all predictors and interactions up to the three-way interaction:

$$\mu_i = \beta_0[\text{Sp}_i] + \beta_P[\text{Sp}_i] P_i + \beta_{\text{endo}}[\text{Sp}_i] endo_i + \beta_{\text{herb}}[\text{Sp}_i] herb_i \\ + \beta_{\text{endo}\times P}[\text{Sp}_i] endo_i P_i + \beta_{\text{herb}\times P}[\text{Sp}_i] herb_i P_i + \beta_{\text{herb}\times\text{endo}}[\text{Sp}_i] herb_i endo_i \\ + \beta_{\text{endo}\times\text{herb}\times P}[\text{Sp}_i] endo_i herb_i P_i + \phi_{\text{site}[i],\text{Sp}_i} + \omega_{\text{source}[i],\text{Sp}_i} + \rho_{\text{plot}[i],\text{Sp}_i}$$

4. **Endophyte + Climate + Herbivory (main effects only):** includes all three predictors but no interactions:

$$\mu_i = \beta_0[\text{Sp}_i] + \beta_{\text{endo}}[\text{Sp}_i] endo_i + \beta_P[\text{Sp}_i] P_i + \beta_{\text{herb}}[\text{Sp}_i] herb_i \\ + \phi_{\text{site}[i],\text{Sp}_i} + \omega_{\text{source}[i],\text{Sp}_i} + \rho_{\text{plot}[i],\text{Sp}_i}$$

In these models, the β coefficients represent species-specific effects. The intercept $\beta_0[\text{Sp}_i]$ corresponds to the baseline value of the vital rate for species Sp_i in the absence of predictors. $\beta_{\text{endo}}[\text{Sp}_i]$ and $\beta_{\text{herb}}[\text{Sp}_i]$ describe the effects of endophyte presence and herbivory, respectively. The linear effect of precipitation is captured by $\beta_P[\text{Sp}_i]$, while interactions are represented by $\beta_{\text{endo}\times P}[\text{Sp}_i]$, $\beta_{\text{herb}\times P}[\text{Sp}_i]$, $\beta_{\text{endo}\times\text{herb}}[\text{Sp}_i]$, and the three-way interaction $\beta_{\text{endo}\times\text{herb}\times P}[\text{Sp}_i]$. All models include hierarchical random effects for site-year, source population, and plot, with species-specific random effects $\phi_{\mathrm{site}[i],\mathrm{Sp}_i} \sim \mathcal{N}(0, \sigma_{\mathrm{site}})$, $\omega_{\mathrm{source}[i],\mathrm{Sp}_i} \sim \mathcal{N}(0, \sigma_{\mathrm{source}})$, and $\rho_{\mathrm{plot}[i],\mathrm{Sp}_i} \sim \mathcal{N}(0, \sigma_{\mathrm{plot}})$.

S.1.2 Supporting Tables

Species	Population	Latitude	Longitude	Notes	Total plants	Unique genotypes	Mean clones per genotype
A. hyemalis	SJC	30.771969	-95.200620	Private property, San Jacinto County, TX	273	201	1.36
A. hyemalis	COHR	29.672833	-96.567517	Roadside collection, Columbus, TX	289	35	8.26
A. hyemalis	LPEF	30.085383	-97.172508	Stengl Lost Pines Field Station, Bastrop, TX	278	41	6.78
P. autumnalis	LA7	30.849660	-93.276772	Roadside population, DeRidder, LA	91	9	10.11
P. autumnalis	LA1	32.568125	-93.932976	Walter P. Jacobs Nature Center, Shreveport, LA	129	7	18.43
P. autumnalis	LA2	31.183040	-92.616969	Kistachie National Forest, LA	253	15	16.87
P. autumnalis	SFAEF	31.492158	-94.772200	Stephen F. Austin Experimental Forest, TX	247	41	6.02
E. virginicus	HUNT	30.741797	-95.474508	Pineywoods Environmental Research Laboratory, Huntsville, TX	91	61	1.49
E. virginicus	RL5	30.471717	-99.783803	Llano River Field Station, Junction, TX	131	91	1.44
E. virginicus	PALM	29.591623	-97.584781	Palmetto State Park, TX	442	195	2.27
E. virginicus	JLP	29.785105	-90.114933	Jean Laffitte Park, LA	176	102	1.73

Table S1: Geographic coordinates and descriptions of source populations for study species. For each source population, the table shows the total number of plants included in the experiment, the number of unique genotypes that these plants comrised, and the average amount of clonal replication per genotype.

Table S2: Summary of endophyte removal treatments by species and population. For each population, N indicates the number of plants that were tested for endophyte status and %E+ indicates the proportion that tested positive. Treatment refers to endophyte removal treatment applied to the seed lineage: Control (no treatment) or Heat (5 days in a drying over at 60°C for *E. virginicus* and *P. autumnalis*, 12 minutes in a 60°C water bath for *A. hyemalis*). For all populations except SFAEF and PALM, heat-treated seed lineages were one or more generations removed from the treatment. For those two populations, plants from heat-treated seeds were used directly in the experiment.

Species	Population	Treatment	N	%E+
A. hyemalis	COHR	Control	84	0.92
A. hyemalis	COHR	Heat	75	0.04
A. hyemalis	LPEF	Control	7	0.57
A. hyemalis	LPEF	Heat	88	0.00
A. hyemalis	SJC	Control	132	1.00
A. hyemalis	SJC	Heat	191	0.05
E. virginicus	HUNT	Control	71	0.61
E. virginicus	HUNT	Heat	38	0.21
E. virginicus	JLP	Control	27	1.00
E. virginicus	JLP	Heat	113	0.02
E. virginicus	PALM	Control	176	0.61
E. virginicus	PALM	Heat	238	0.71
E. virginicus	RL5	Control	68	0.43
E. virginicus	RL5	Heat	74	0.73
P. autumnalis	LA1	Control	5	0.20
P. autumnalis	LA1	Heat	4	0.00
P. autumnalis	LA2	Control	10	0.80
P. autumnalis	LA2	Heat	3	0.33
P. autumnalis	LA7	Control	5	1.00
P. autumnalis	LA7	Heat	11	0.00
P. autumnalis	SFAEF	Control	23	0.35
P. autumnalis	SFAEF	Heat	38	0.26

Table S3: Common garden sites used for reciprocal transplant and environmental manipulation studies.

Site	Code	Latitude	Longitude
University of Louisiana Ecology Center	LAF	30.304773	-92.012401
Center for Biological Field Studies	HUN	30.742815	-95.476881
TAMU Ecology and Natural Resource Teaching Area	COL	30.569302	-96.366324
UT Stengl Lost Pines Field Station	BAS	30.092343	-97.172442
Private property in Kerrville, TX	KER	30.028437	-99.142888
Brackenridge Field Laboratory	BFL	30.284866	-97.780675
TAMU Sonora Station	SON	30.273467	-100.561463

Table S4: Comparison of candidate models (quadratic vs. linear) for each vital rate based on leave-one-out cross-validation (LOOCV). ELPD difference is relative to the best model.

Vital rate	Model type	ELPD difference	SE difference
Survival	Quadratic	0	0.00
Survival	Linear	-0.055	1.09
Growth	Quadratic	0	0.00
Growth	Linear	-0.187	1.33
Inflorescence	Quadratic	0	0.00
Inflorescence	Linear	-1.510	1.43
Spikelet	Quadratic	0	0.00
Spikelet	Linear	-0.849	1.32

Table S5: Posterior summaries of Δ ($E^{+} - E^{-}$) for survival across precipitation quantiles (mm) for each species and herbivore treatment. Positive median values indicate beneficial effects of endophytes, negative medians indicate costly effects. Posterior probabilities $P(\Delta > 0)$ quantify the strength of evidence: values near 1 indicate strong support for a benefit, values near 0 indicate strong support for a cost, and values near 0.5 indicate little or no consistent effect. Herbivore treatment: Exclusion = herbivores excluded, Access = herbivores had access.

Species	Herbivore treatment	Precipitation	Median Δ	Lower 90%	Upper 90%	$P(\Delta > 0)$
A. hyemalis	Access	543	-0.034	-0.282	0.061	0.189
A. hyemalis	Access	786	-0.029	-0.183	0.057	0.225
A. hyemalis	Access	1285	0.000	-0.076	0.071	0.496
A. hyemalis	Access	1858	0.035	-0.049	0.163	0.775
A. hyemalis	Access	3231	0.079	-0.052	0.353	0.842
A. hyemalis	Exclusion	543	-0.077	-0.348	0.000	0.052
A. hyemalis	Exclusion	786	-0.068	-0.240	0.002	0.057
A. hyemalis	Exclusion	1285	-0.027	-0.113	0.038	0.233
A. hyemalis	Exclusion	1858	0.023	-0.071	0.145	0.677
A. hyemalis	Exclusion	3231	0.085	-0.047	0.335	0.863
E. virginicus	Access	543	0.010	-0.141	0.208	0.566
E. virginicus	Access	786	0.019	-0.128	0.184	0.598
E. virginicus	Access	1285	0.032	-0.096	0.161	0.663
E. virginicus	Access	1858	0.037	-0.096	0.172	0.679
E. virginicus	Access	3231	0.031	-0.151	0.220	0.628
E. virginicus	Exclusion	543	-0.168	-0.404	-0.010	0.040
E. virginicus	Exclusion	786	-0.143	-0.309	0.016	0.068
E. virginicus	Exclusion	1285	-0.005	-0.119	0.112	0.473
E. virginicus	Exclusion	1858	0.080	-0.027	0.218	0.896
E. virginicus	Exclusion	3231	0.092	0.006	0.307	0.966
P. autumnalis	Access	543	-0.001	-0.082	0.008	0.231
P. autumnalis	Access	786	-0.006	-0.132	0.017	0.195
P. autumnalis	Access	1285	-0.057	-0.192	0.036	0.147
P. autumnalis	Access	1858	-0.022	-0.177	0.087	0.328
P. autumnalis	Access	3231	0.000	-0.105	0.103	0.533
P. autumnalis	Exclusion	543	0.000	-0.018	0.065	0.620
P. autumnalis	Exclusion	786	0.002	-0.035	0.097	0.647
P. autumnalis	Exclusion	1285	0.025	-0.072	0.150	0.705
P. autumnalis	Exclusion	1858	0.016	-0.105	0.167	0.628
P. autumnalis	Exclusion	3231	0.000	-0.107	0.116	0.531

Table S6: Posterior summaries of Δ ($E^{+} - E^{-}$) for growth across precipitation quantiles (mm) for each species and herbivore treatment. Positive median values indicate beneficial effects of endophytes, negative medians indicate costly effects. Posterior probabilities $P(\Delta > 0)$ quantify the strength of evidence: values near 1 indicate strong support for a benefit, values near 0 indicate strong support for a cost, and values near 0.5 indicate little or no consistent effect.

Species	Herbivore treatment	Precipitation	Median Δ	Lower 90%	Upper 90%	$P(\Delta > 0)$
A. hyemalis	Access	543	0.013	-0.610	0.651	0.514
A. hyemalis	Access	786	0.152	-0.285	0.596	0.718
A. hyemalis	Access	1285	0.341	0.069	0.609	0.981
A. hyemalis	Access	1858	0.482	0.167	0.788	0.994
A. hyemalis	Access	3231	0.691	0.125	1.252	0.977
A. hyemalis	Exclusion	543	-0.201	-0.725	0.316	0.261
A. hyemalis	Exclusion	786	-0.053	-0.418	0.300	0.403
A. hyemalis	Exclusion	1285	0.140	-0.096	0.377	0.838
A. hyemalis	Exclusion	1858	0.288	-0.014	0.584	0.941
A. hyemalis	Exclusion	3231	0.510	-0.019	1.033	0.944
E. virginicus	Access	543	0.552	-0.021	1.126	0.943
E. virginicus	Access	786	0.455	0.021	0.880	0.959
E. virginicus	Access	1285	0.323	0.045	0.596	0.972
E. virginicus	Access	1858	0.223	-0.017	0.465	0.936
E. virginicus	Access	3231	0.076	-0.296	0.438	0.631
E. virginicus	Exclusion	543	-0.087	-0.609	0.432	0.387
E. virginicus	Exclusion	786	-0.034	-0.422	0.345	0.441
E. virginicus	Exclusion	1285	0.036	-0.196	0.266	0.606
E. virginicus	Exclusion	1858	0.090	-0.096	0.278	0.787
E. virginicus	Exclusion	3231	0.171	-0.120	0.470	0.829
P. autumnalis	Access	543	-0.385	-1.094	0.324	0.188
P. autumnalis	Access	786	-0.256	-0.753	0.248	0.198
P. autumnalis	Access	1285	-0.083	-0.339	0.179	0.300
P. autumnalis	Access	1858	0.048	-0.188	0.282	0.635
P. autumnalis	Access	3231	0.239	-0.247	0.723	0.799
P. autumnalis	Exclusion	543	0.744	0.078	1.438	0.967
P. autumnalis	Exclusion	786	0.468	-0.001	0.947	0.950
P. autumnalis	Exclusion	1285	0.095	-0.156	0.352	0.734
P. autumnalis	Exclusion	1858	-0.184	-0.429	0.067	0.109
P. autumnalis	Exclusion	3231	-0.604	-1.091	-0.106	0.023

Table S7: Posterior summaries of Δ ($E^{+} - E^{-}$) for inflorescences across precipitation quantiles (mm) for each species and herbivore treatment. Positive median values indicate beneficial effects of endophytes, negative medians indicate costly effects. Posterior probabilities $P(\Delta > 0)$ quantify the strength of evidence: values near 1 indicate strong support for a benefit, values near 0 indicate strong support for a cost, and values near 0.5 indicate little or no consistent effect.

Species	Herbivore treatment	Precipitation	Median Δ	Lower 90%	Upper 90%	$P(\Delta > 0)$
A. hyemalis	Access	543	-0.255	-1.127	0.302	0.179
A. hyemalis	Access	786	-0.261	-1.077	0.409	0.226
A. hyemalis	Access	1285	-0.001	-0.797	0.800	0.499
A. hyemalis	Access	1858	0.718	-0.486	2.578	0.832
A. hyemalis	Access	3231	3.915	-0.863	18.461	0.899
A. hyemalis	Exclusion	543	-0.286	-1.282	0.370	0.187
A. hyemalis	Exclusion	786	-0.260	-1.163	0.503	0.254
A. hyemalis	Exclusion	1285	0.152	-0.711	1.103	0.623
A. hyemalis	Exclusion	1858	1.126	-0.433	3.585	0.877
A. hyemalis	Exclusion	3231	5.224	-0.833	23.097	0.912
E. virginicus	Access	543	-0.002	-0.534	0.691	0.498
E. virginicus	Access	786	0.012	-0.416	0.529	0.522
E. virginicus	Access	1285	0.040	-0.275	0.390	0.585
E. virginicus	Access	1858	0.068	-0.258	0.422	0.640
E. virginicus	Access	3231	0.122	-0.555	0.989	0.629
E. virginicus	Exclusion	543	-0.385	-1.582	0.555	0.210
E. virginicus	Exclusion	786	-0.335	-1.253	0.490	0.224
E. virginicus	Exclusion	1285	-0.187	-0.838	0.431	0.295
E. virginicus	Exclusion	1858	-0.001	-0.523	0.527	0.498
E. virginicus	Exclusion	3231	0.424	-0.477	1.810	0.788
P. autumnalis	Access	543	0.015	-0.237	1.502	0.655
P. autumnalis	Access	786	0.057	-0.132	1.162	0.750
P. autumnalis	Access	1285	0.235	-0.006	1.174	0.944
P. autumnalis	Access	1858	0.579	0.072	3.125	0.996
P. autumnalis	Access	3231	1.681	-0.124	43.735	0.925
P. autumnalis	Exclusion	543	0.377	0.012	11.777	0.985
P. autumnalis	Exclusion	786	0.602	0.061	6.436	0.994
P. autumnalis	Exclusion	1285	1.050	0.263	4.237	1.000
P. autumnalis	Exclusion	1858	1.461	0.236	7.731	1.000
P. autumnalis	Exclusion	3231	1.139	-2.399	43.360	0.810

Table S8: Posterior summaries of Δ ($E^+ - E^-$) for spikelets across precipitation quantiles (mm) for each species and herbivore treatment. Positive median values indicate beneficial effects of endophytes, negative medians indicate costly effects. Posterior probabilities $P(\Delta > 0)$ quantify the strength of evidence: values near 1 indicate strong support for a benefit, values near 0 indicate strong support for a cost, and values near 0.5 indicate little or no consistent effect.

Species	Herbivore treatment	Precipitation	Median Δ	Lower 90%	Upper 90%	$P(\Delta > 0)$
E. virginicus	Access	543	0.154	-5.828	7.167	0.519
E. virginicus	Access	786	0.220	-4.461	5.440	0.533
E. virginicus	Access	1285	0.341	-2.960	3.787	0.570
E. virginicus	Access	1858	0.419	-2.755	3.708	0.595
E. virginicus	Access	3231	0.572	-5.092	6.550	0.574
E. virginicus	Exclusion	543	-2.568	-8.701	3.243	0.208
E. virginicus	Exclusion	786	-1.971	-6.772	2.814	0.231
E. virginicus	Exclusion	1285	-0.866	-4.104	2.452	0.321
E. virginicus	Exclusion	1858	0.196	-2.261	2.726	0.554
E. virginicus	Exclusion	3231	2.168	-1.414	7.037	0.842
P. autumnalis	Access	543	-0.548	-6.176	3.403	0.336
P. autumnalis	Access	786	-0.528	-4.719	2.868	0.352
P. autumnalis	Access	1285	-0.237	-2.653	2.158	0.424
P. autumnalis	Access	1858	0.396	-1.674	2.658	0.627
P. autumnalis	Access	3231	2.179	-4.709	12.650	0.724
P. autumnalis	Exclusion	543	2.840	-1.338	13.112	0.891
P. autumnalis	Exclusion	786	3.170	-0.535	10.293	0.924
P. autumnalis	Exclusion	1285	3.378	0.635	7.134	0.979
P. autumnalis	Exclusion	1858	3.123	0.477	6.660	0.976
P. autumnalis	Exclusion	3231	1.632	-7.556	12.272	0.643

S.1.3 Supporting Figures

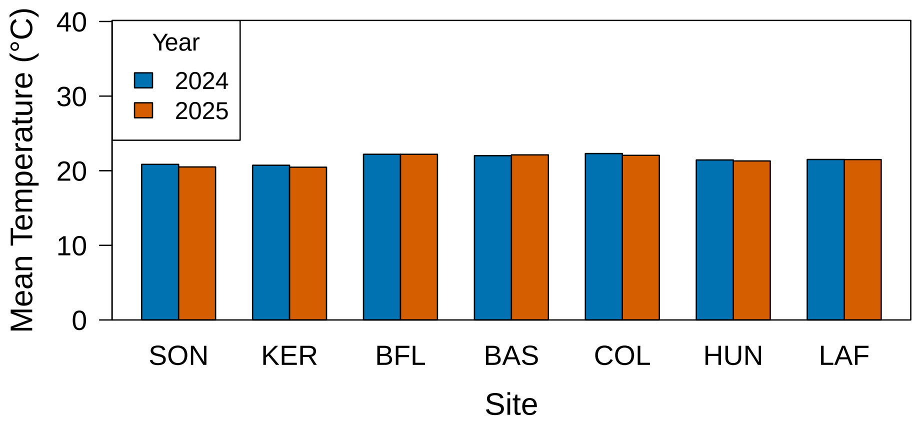

Fig S1: Annual temperature (°C) for 2024 and 2025 census. Sites are arranged from west to east based on longitude.

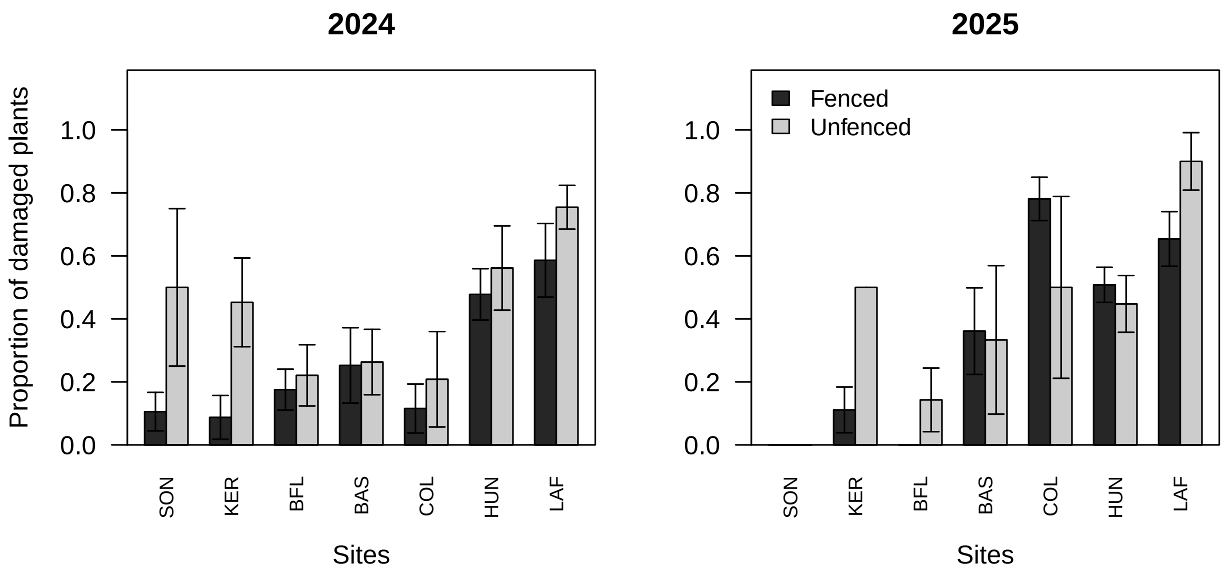

Fig S2: Proportion of plants damaged by herbivory across sites in 2024 and 2025. Barplots show fenced (dark) versus unfenced (grey) plots for each site (ordered west to east). Error bars indicate standard errors across plots within each site. Panels show results for 2024 (left) and 2025 (right).

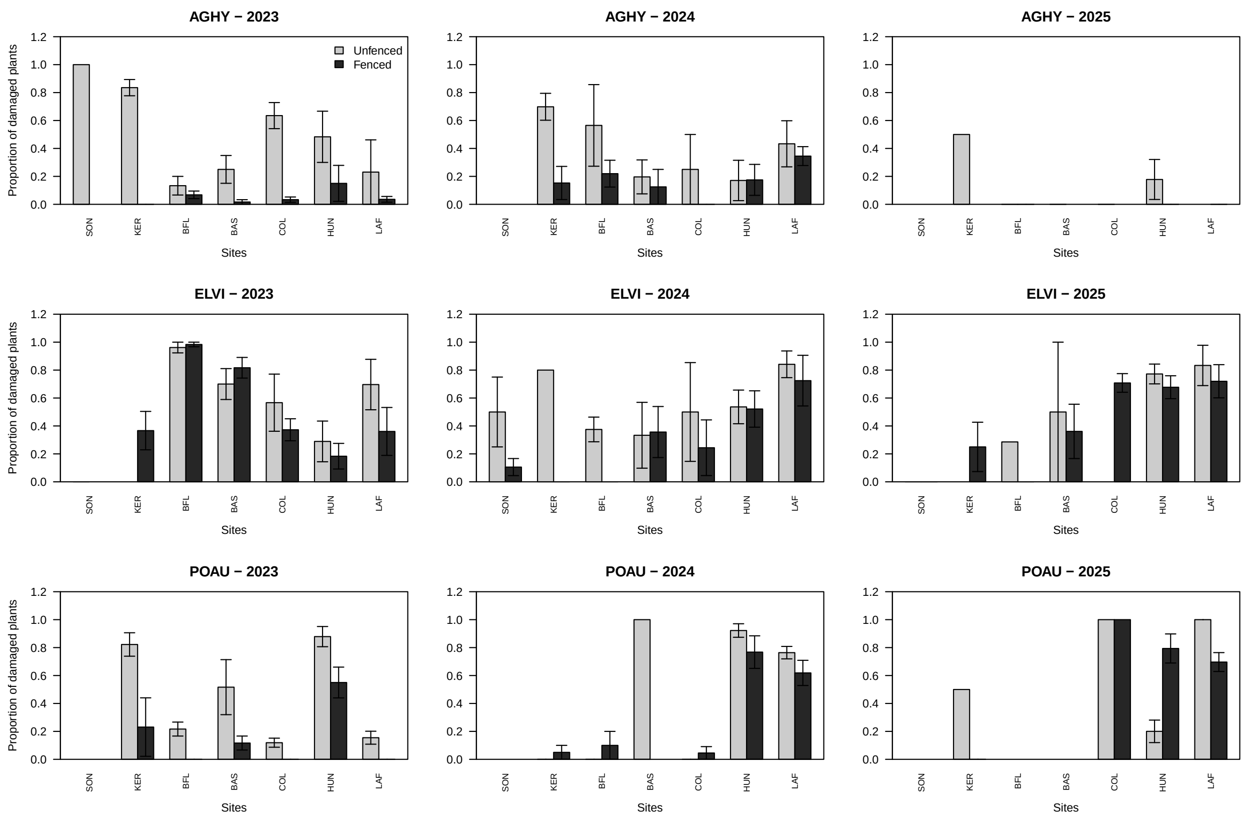

Fig S3: Proportion of plants damaged by herbivory across sites for three species (*Agrostis hyemalis*, *Elymus virginicus*, *Poa autumnalis*) over the years 2023–2025. Barplots show fenced (grey80) versus unfenced (grey15) proportion of damage for each site (ordered west to east: SON, KER, BFL, BAS, COL, HUN, LAF). Error bars indicate standard errors across plots within each site. Panels are arranged by species (rows) and year (columns), with the y-axis labeled only for the first column for clarity.

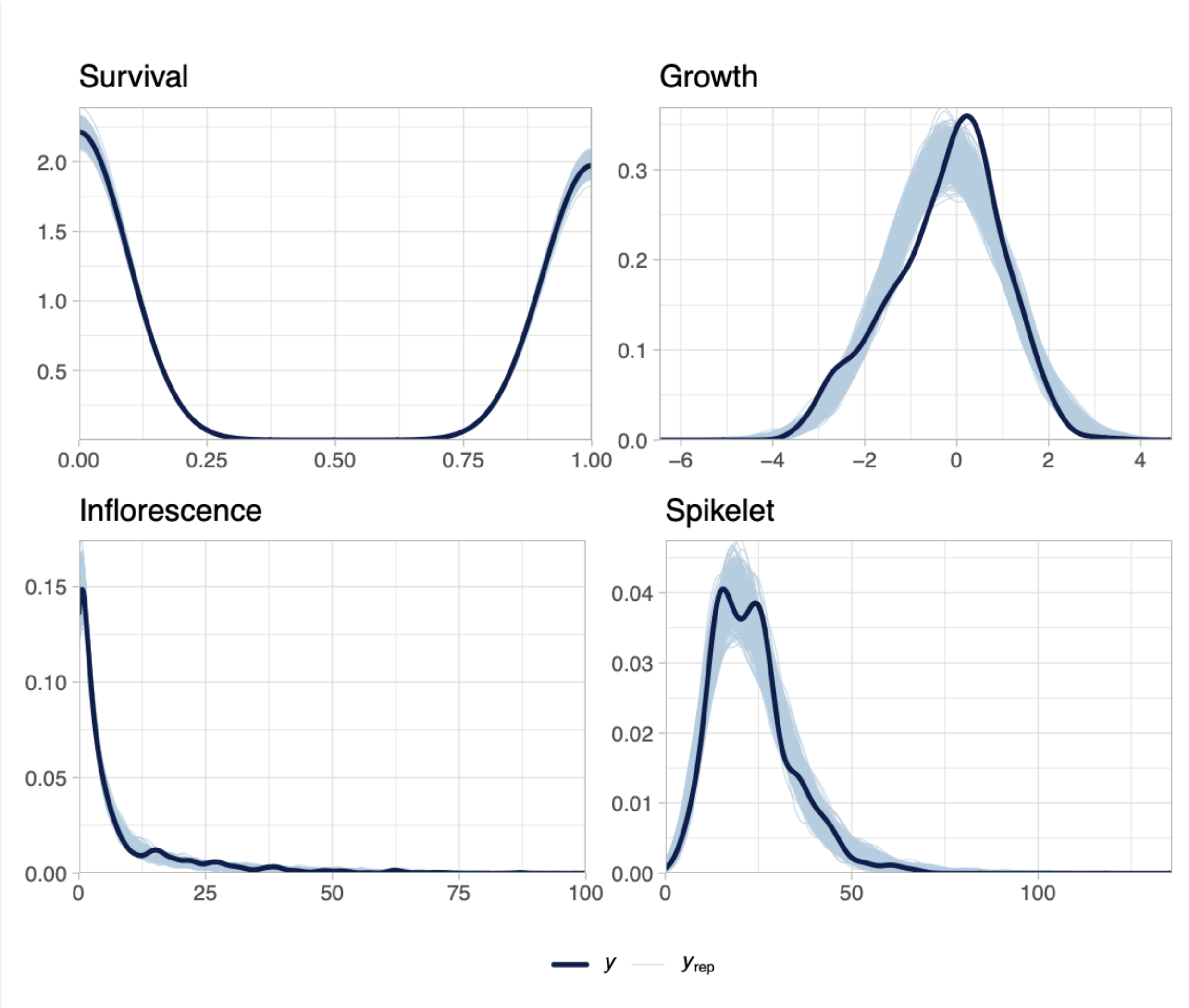

Fig S4: Posterior predictive checks for each vital rate. Dark blue lines show observed data (y), and light blue lines show replicated datasets generated from posterior predictions (y_{rep}). The strong overlap across survival, growth, flowering, and spikelet production indicates adequate model fit for all vital rates.

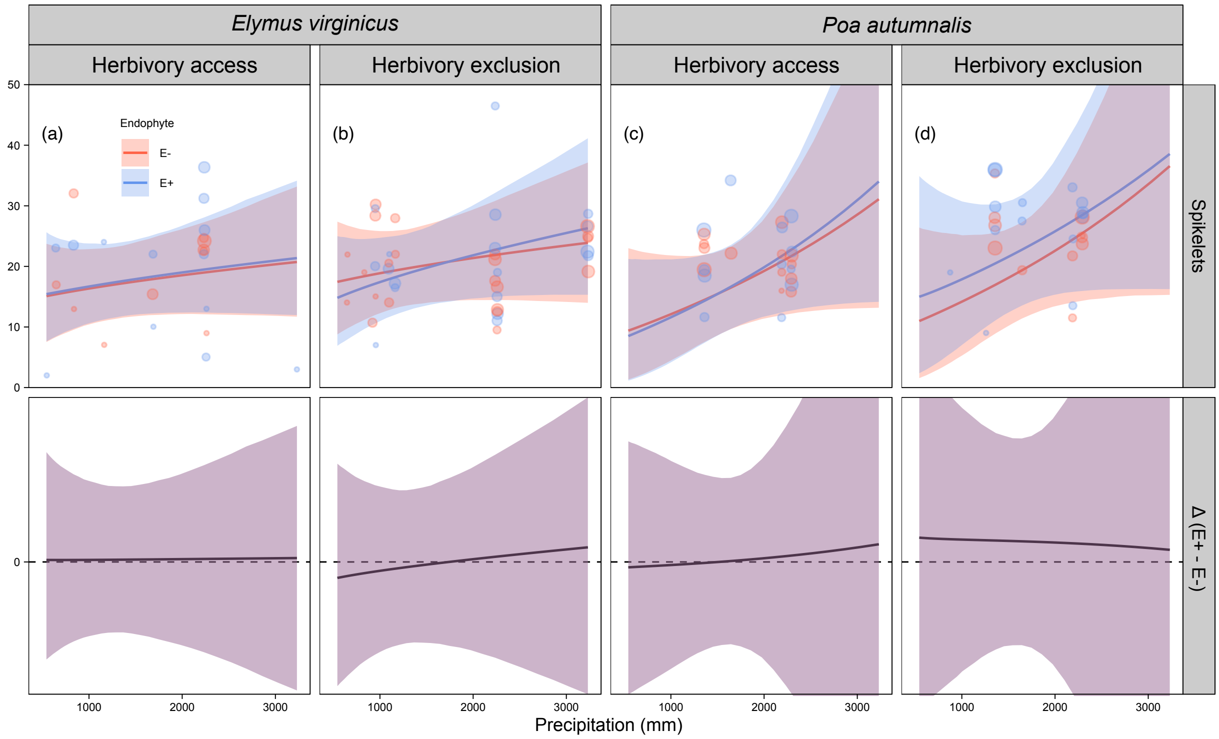

Fig S5: Predicted number of spikelets (upper panels) and endophyte effect ($\Delta = E^{+} - E^{-}$, bottom panels) across precipitation gradients. Lines show posterior means for E^{+} (blue) and E^{-} (red), with shading indicating 90% credible intervals. Points represent mean observed spikelets per plot, slightly jittered for visualization. Spikelets were measured for only two species (see Methods for details). The dashed line at $\Delta = 0$ indicates no endophyte effect. Panels are organized by species and herbivory treatment (Fenced, Unfenced).

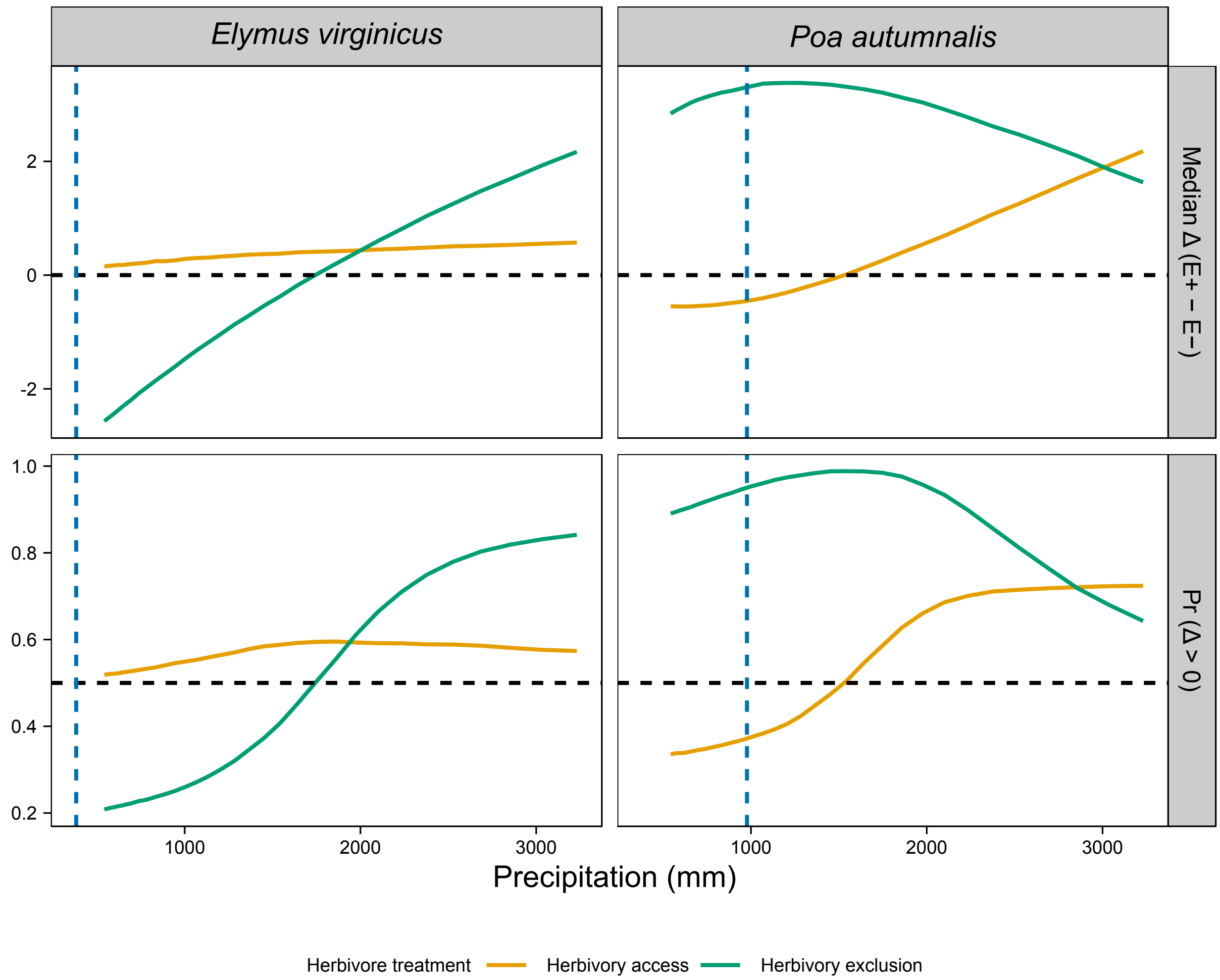

Fig S6: Effects of precipitation and endophyte presence on spikelet (reproductive output) for three cool-season grass species. Median Δ ($E^{+} - E^{-}$) and posterior probability $\Pr(\Delta > 0)$ are shown across precipitation quantiles. Herbivore exclusion treatments are indicated by color (Fenced = herbivores excluded, Unfenced = herbivores had access). Horizontal dashed lines indicate $\Delta = 0$ for medians and $\Pr(\Delta > 0) = 0.5$ for probabilities. Facets separate metrics and species.